

Watershed Delineation and Prioritization Using ArcMap 10.8

Abstract:

Watershed delineation and prioritization play a crucial role in sustainable water resource management and soil conservation planning. This study utilizes ArcMap 10.8.2 for watershed delineation based on SRTM 30-meter Digital Elevation Model (DEM) and assigns priority rankings to sub-watersheds using various morphometric parameters. The final priority map provides insights for decision-makers to focus conservation efforts efficiently.

1. Introduction:

Watershed delineation is the process of defining the boundary of a drainage basin using elevation data. Prioritization of sub-watersheds based on their susceptibility to erosion and runoff is essential for effective resource management. In this study, ArcMap 10.8 is used for delineating the watershed and prioritizing sub-watersheds using morphometric analysis. The final output includes a priority table and a sub-watershed-wise priority map.

2. Data Used:

- Satellite Data: SRTM 30-meter Digital Elevation Model (DEM)
- Software Used: ArcMap 10.8.2
- Other Data: Drainage network, watershed boundary, and field observation records (if available)

3. Methodology:

3.1 Watershed Delineation

The following steps are performed in ArcMap 10.8.2 for watershed delineation:

1. DEM Preprocessing:

- Fill sinks in the DEM to remove artificial depressions.
- Generate flow direction and flow accumulation raster.

2. Stream Network Extraction:

- Define a threshold for flow accumulation to extract streams.
- Convert raster stream network to vector format.

3. Watershed Delineation:

- Identify pour points and use them to delineate sub-watersheds.
- Generate sub-watershed boundaries

3.2 Morphometric Analysis and Prioritization

Various morphometric parameters are calculated for prioritization:

- Linear Parameters: Stream length (Lu), bifurcation ratio (Rb), drainage density (Dd)
- Areal Parameters: Drainage texture (Dt), form factor (Rf), circularity ratio (Rc)
- Relief Parameters: Basin relief (H), relief ratio (Rh)

The computed values are compiled into a tabular format for better understanding.

Sub-Watershed	Stream Order (U)	Stream Number (Nu)	Stream Length (Lu) (m)	Drainage Density (Dd)	Stream Frequency (Fs)	Basin Area (A) (sq.m)	Basin Perimeter (P) (m)	Form Factor (Ff)	Circularity Ratio (Rc)
SW 1	5	236	2.9648	112.0695521	8920.808921	0.026455	1.298	0.015702123	0.197218668
SW 2	6	383	5.1487	107.0838793	7965.724507	0.048081	1.976	0.012314013	0.154664004
SW 3	5	139	1.6845	110.4735047	9115.949633	0.015248	0.8646	0.020397757	0.256195832
SW 4	5	107	1.389	106.0791202	8171.681686	0.013094	0.8908	0.016501063	0.20725335
SW 5	5	355	4.34	106.9782346	8750.523799	0.040569	1.3464	0.022379279	0.281083748
SW 6	4	141	1.6984	103.3027188	8576.120674	0.016441	0.7508	0.02916619	0.366327347
SW 7	4	324	3.8336	104.2504011	8810.812281	0.036773	1.439	0.017758549	0.223047373
SW 8	4	94	1.0336	98.80508556	8985.75662	0.010461	0.6358	0.025878086	0.325028762

Basin Relief (H) (m)	Relief Ratio (Rh)	Ruggedness Number (Rn)	Rank (Dd)	Rank (Fs)	Rank (Rh)	Rank (Rn)	Compound Parameter (Cp)	Priority
781	601.6949153	87526.32017	8	6	5	5	6	Low
994	503.0364372	106441.376	6	1	8	8	5.75	Low
849	981.9569743	93792.00551	7	8	3	3	5.25	Medium
837	939.6048496	88788.22361	4	2	2	4	3	High
973	722.667855	104089.8223	5	4	7	6	5.5	Medium

1037	1381.1933 94	107124.91 94	2	3	4	2	2.75	High
1038	721.33425 99	108211.91 64	3	5	6	7	5.25	Medium
991	1558.6662 47	97915.839 79	1	7	1	1	2.5	High

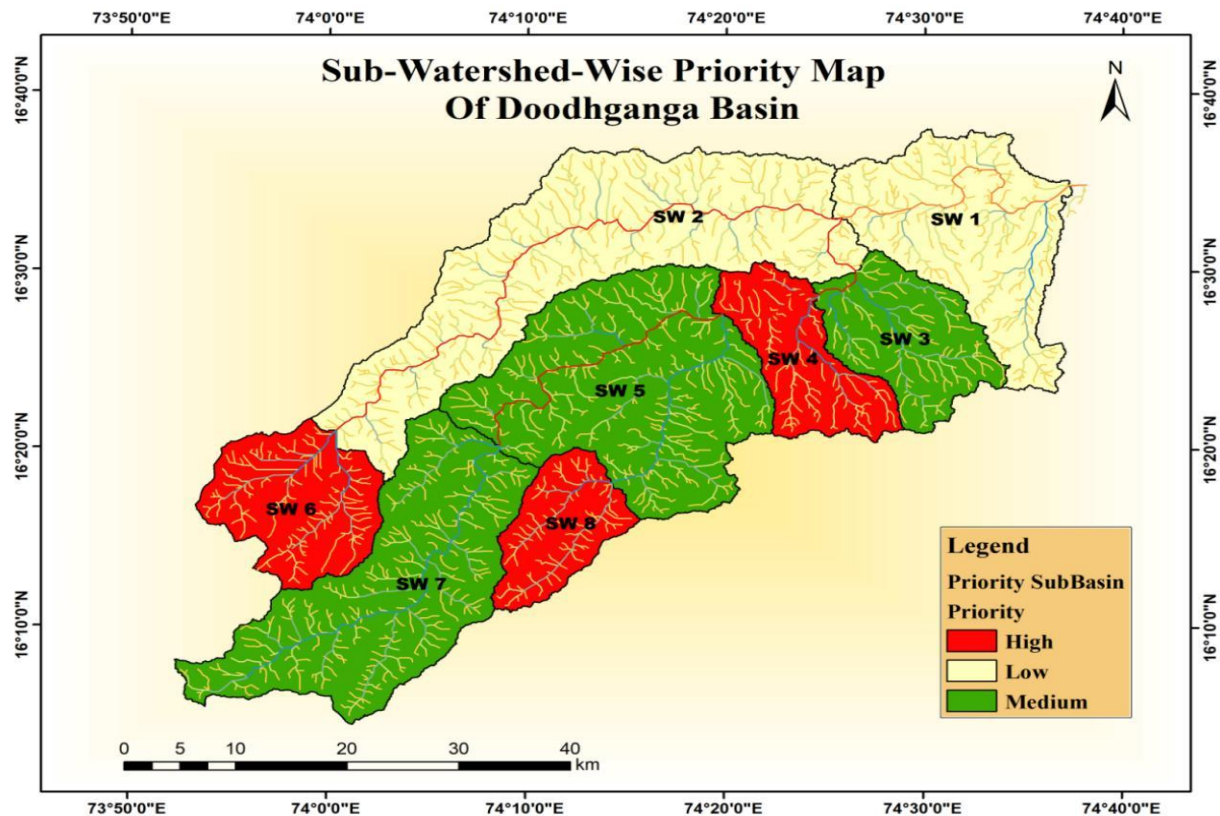
Table 1: Morphometric Parameter Values and Prioritization Scores

3.3 Priority Ranking

- Each sub-watershed is ranked based on morphometric analysis.
- A priority score is assigned based on the average ranking.
- Higher values indicate higher susceptibility to erosion, requiring urgent conservation measures.

4. Results and Discussion:

- The priority table for sub-watersheds presents the rankings based on calculated morphometric parameters.
- The final sub-watershed-wise priority map highlights high-priority areas requiring immediate intervention.



5. Conclusion:

This study successfully delineated watersheds and prioritized them based on morphometric analysis using ArcMap 10.8 and SRTM 30m DEM. The generated priority map serves as a valuable tool for soil and water conservation planning. Future studies can integrate additional hydrological models and land use data for more precise prioritization.